

Enhanced Detection of Age-Related and Cognitive Declines Using Automated Hippocampal-to-Ventricle Ratio in Alzheimer's Patients

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	Clinical Label		
	CH, N: 501	MCI, N: 819	AD, N: 321
Sex (F) ¹	260 (52%)	336 (41%)	146 (45%)
Age (years) ²	74.3 (5.7)	72.9 (7.6)	74.8 (7.8)
Education (years) ²	16.3 (2.7)	15.9 (2.8)	15.2 (3.0)
ADAS-13 ²	9.3 (4.4)	16.6 (6.8)	29.9 (8.0)
Missing ³	2	9	11
RAVLT-learning ²	5.8 (2.3)	4.1 (2.6)	1.8 (1.8)
Missing ³	4	4	4

¹N (%). ²Mean (SD). ³N of subjects without cognitive data.

TABLE 1 – Baseline demographics and cognitive data.
Sex distribution, age, education, and scores for ADAS-13 and the learning subtest of the RAVLT for the CH, MCI and AD groups of the ADNI baseline dataset.

	Clinical Label		
	CH, N: 501	MCI, N: 819	AD, N: 321
MALF , N: 1,617			
Hippocampus ¹	4.37 (0.39)	4.19 (0.42)	4.06 (0.41)
HVR ¹	0.69 (0.03)	0.67 (0.04)	0.65 (0.03)
Failures ²	3	8	13
NLPB , N: 1,627			
Hippocampus ¹	4.14 (0.49)	3.79 (0.61)	3.38 (0.53)
HVR ¹	0.67 (0.08)	0.62 (0.10)	0.54 (0.11)
Failures ²	2	9	3
CNN , N: 1,639			
Hippocampus ¹	4.18 (0.55)	3.73 (0.71)	3.16 (0.65)
HVR ¹	0.64 (0.08)	0.58 (0.11)	0.49 (0.11)
Failures ²	-	1	1
FreeSurfer v4.3 & v5.1 , N: 1,414			
Hippocampus ¹	5.16 (0.71)	4.63 (0.92)	3.90 (0.72)
Missing ³	47	122	58
FreeSurfer v6.0 , N: 1,616			
Hippocampus ¹	5.04 (0.68)	4.50 (0.90)	3.85 (0.71)
HVR ¹	0.84 (0.08)	0.77 (0.12)	0.67 (0.13)
Failures ²	8	9	8

¹ Mean (SD). ² N of failed segmentations. ³ N of unreported cases from ADNI.

TABLE 2 – Summary of hippocampal volumes (HCvol), hippocampal-to-ventricle ratio (HVR), and segmentation failures.

Mean and standard deviation of the obtained HCvol and calculated HVR, along with the number of segmentation failures for each automatic segmentation method. *Note:* For FreeSurfer versions 4.3 and 5.1, missing data from ADNI are reported instead of explicit algorithm failures. Due to the lack of ventricular volume data, HVR was not calculated for these versions of FreeSurfer.